*.tem

Purpose

A `*.tem` file is a temperature time-series file with the same record layout as a `*.tmp`. The `.tem` extension is used by the reference datasets shipped with SWAT+ (for example `refdata/Ames_sub1/ames.tem`) and by some external preprocessors. There is no separate reader for `.tem`. The temperature reader at `src/cli_tmeas.f90` opens whatever filename is supplied; the extension is not inspected.

Source

- Reader: `src/cli_tmeas.f90`
- Type definition: `src/climate_module.f90`, `type climate_measured_data`

The reader is reached only when the filename appears in `tmp.cli`. The `tgage` slot in `weather-sta.cli` is matched against the filename listed in `tmp.cli` via the `tmp_n` lookup array, so both files must agree on the string (including the `.tem` extension if used).

Format

Identical to a daily `*.tmp` file. List-directed reads, whitespace-separated.

- Line 1: title (skipped).
- Line 2: column header (skipped). Conventionally `NBYR  TSTEP  LAT  LONG  ELEV`.
- Line 3: station header (`nbyr`, `tstep`, `lat`, `long`, `elev`).
- Line 4 onward: one record per day with year, Julian day, daily maximum temperature, daily minimum temperature.

See `*.tmp` for the field tables.

Example

`refdata/Ames_sub1/ames.tem` (excerpt):

```
AME.Tmp
 NBYR    TSTEP      LAT      LONG     ELEV
   121        0    42.04   -93.89   316
1900    1   -99 -99
1900    2   -99 -99
...
1983    185 32.2    10.6
1983    186 26.7    11.1
```

Related files

- `*.tmp`. Same content, alternate extension. The reader is shared.
- `tmp.cli`. Lists each temperature filename (with whatever extension).
- `weather-sta.cli`. The `tgage` column must match a filename entry in `tmp.cli`.